

Region Affinities

Summary

Plain-Language Overview

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Companion documents: Technical Appendix · Methods Primer · Data Appendix · Region Reference — Identity · Region Reference — Terroir

Abstract. *Wine regions can be classified in two fundamentally different ways: by physical terroir (geology, climate, topography) and by cultural identity (narrative, ambition, historical character). This study constructs independent clustering solutions for both classification systems across 59 wine regions spanning 16 countries, and tests whether the two solutions agree. Identity is encoded through six expert-scored dimensions ($k = 6$); terroir is encoded through TF-IDF vectorisation of structured terroir profiles ($k = 7$, fully model-derived). Agreement is measured using the Adjusted Rand Index and a chi-squared test of independence. The result is unambiguous: $\text{ARI} = 0.023$, $\chi^2 = 38.78$ ($p = 0.131$). The two classifications are statistically independent. A region's physical terroir is essentially uninformative about its cultural identity cluster, and vice versa. The Region Affinities tool exposes both classification systems through interactive similarity networks and a side-by-side comparison view.*

1. Introduction

The dominant framework for understanding wine regions is geographic. Appellation systems — from Burgundy's grands crus to Napa's sub-AVAs — are organised by land. The implicit assumption is that physical terroir is the primary organising principle of regional character: regions in similar climatic and geological settings should develop similar identities. Region Affinities tests this assumption directly.

Two independent classification systems are applied to the same set of 59 wine regions: one grounded in physical terroir features (climate, soils, principal varieties, winemaking, production structure, historical position), and one grounded in six anthropological identity dimensions (interiority, struggle, tradition, individuality, urgency, earthliness). Both systems produce a clustering of the 59 regions; the central question is whether the two clusterings agree.

The finding is that they do not. Across all standard measures of cluster agreement, the two classifications behave as if they were drawn independently. This is not an argument against terroir as an organising principle for what wines are like; it is evidence that the cultural identity of a wine region operates as a second system, in parallel to geography, governed by its own logic. The remainder of this document describes the data, methods, results, robustness checks, and conclusions that support this finding.

2. Data

2.1 Region Universe

The dataset comprises 59 wine regions selected for global significance, regional diversity, and representation across the canonical literature of wine. The selection spans 16 countries across both Old World (39 regions, 10 countries) and New World (20 regions, 6 countries) classifications. Major viticultural traditions are represented: France (10 regions), Italy (10), USA (9), Germany (5), Spain (4), Austria (4), Australia (3), Greece (2), and 8 other countries with one or two regions each.

2.2 Identity Encoding (Layer 1)

Each region is encoded as a six-dimensional vector of integer scores in $\{-2, -1, 0, +1, +2\}$. The six dimensions, denoted D1 through D6, are:

Code	Dimension	Interpretation
D1	Interiority $\leftrightarrow$ Exteriority	Interior, place-defined vs. outward-facing, market-oriented
D2	Struggle $\leftrightarrow$ Ease	Extreme difficulty, endurance vs. pleasure, comfort
D3	Tradition $\leftrightarrow$ Reinvention	Deep custodial tradition vs. radical reinvention
D4	Individual $\leftrightarrow$ Collective	Strongly individual, solitary vs. strongly collective
D5	Urgency $\leftrightarrow$ Timelessness	Urgent, present-focused vs. timeless, eternal
D6	Earthly $\leftrightarrow$ Transcendent	Deeply grounded vs. transcendent, spiritual

The identity vector for each region (the D-score profile) is derived from a roughly 300-word anthropological identity narrative for that region. Narratives are written without reference to terroir, grape varieties, or winemaking technique: they describe cultural character only. The full set of 59 narratives is provided in the companion document *Region Reference — Identity*.

2.3 Terroir Encoding (Layer 2)

Each region is also described by a structured terroir profile covering six factual fields: climate, soils, principal varieties, winemaking, production structure, and historical position. These profiles are vectorised using term frequency–inverse document frequency (TF-IDF) with bigram features (`ngram_range = (1, 2)`, `max_features = 800`, `sublinear_tf = True`), then reduced via principal component analysis to 10 components retaining 32.2% of variance. The resulting low-dimensional terroir vectors are entirely model-derived; no expert scoring is applied at this layer. The full set of 59 terroir profiles is provided in *Region Reference — Terroir*.

2.4 Independence of the Two Layers

The critical property of this design is that the two encoding layers share no input. The identity layer reads only the cultural narrative; the terroir layer reads only the factual terroir profile. No region appears in both encodings under the same features, and no information leaks between the two layers. This independence at the input stage is what makes the downstream agreement test meaningful.

3. Methodology

The methodology consists of three components: independent clustering of the two layers, similarity-based nearest-neighbour graph construction for visualisation, and a formal test of agreement between the two clusterings.

3.1 Identity Clustering

The 59 D-score profiles are standardised using `StandardScaler`, projected onto their full six principal components (retaining 100% of variance), and partitioned using K-means with $k = 6$, `n_init = 20`, `random_state = 42`. The choice of $k = 6$ is informed by silhouette analysis across $k \in \{3, \dots, 10\}$ and by qualitative interpretability of the resulting clusters.

3.2 Terroir Clustering

The 59 TF-IDF terroir vectors are projected onto their first 10 principal components and partitioned using K-means with $k = 7$, `n_init = 20`, `random_state = 42`. The choice of $k = 7$ is fully model-derived: it maximises the silhouette score across $k \in \{3, \dots, 10\}$ with no manual override.

3.3 Similarity Networks

For each layer, pairwise cosine similarity is computed between region vectors, and a k -nearest-neighbour graph with $k = 5$ is constructed for visualisation. Each node is a region; each edge connects a region to one of its five most similar peers in that layer. The two networks — one identity, one terroir — are presented side by side in the Dual Networks view of the tool.

3.4 Cluster Agreement: ARI and Chi-Squared

Agreement between the identity clustering and the terroir clustering is measured using the Adjusted Rand Index (ARI). ARI counts the pairs of regions that are placed in the same cluster (or in different clusters) by both solutions, normalised so that random labellings score zero in expectation and identical labellings score one. As a complementary test, the 6×7 contingency table of identity-cluster membership against terroir-cluster membership is evaluated using Pearson’s chi-squared test of independence. Full mathematical treatment in *Technical Appendix* (Sections A.4–A.6).

4. Results

4.1 Identity Clusters

The identity clustering yields six clusters, ranging in size from 5 to 15 regions:

Cluster	Size	Defining Character
The Moderates	15	Balanced across all dimensions; equilibrium over extremity
New World Reinvention	12	Constructed identities, urgency, disruption
Old World Interior	10	Tradition-bound, place-defined, custodial
Against the Odds	10	Defined by struggle, extreme terrain, marginalisation
Old World Exterior	7	Institutional, market-facing, tradition-holding
Outward Ease	5	Outward-facing, pleasure-oriented, accessible

Silhouette score for the identity clustering is 0.303, indicating well-separated structure for social-science data.

4.2 Terroir Clusters

The terroir clustering yields seven clusters, ranging in size from 3 to 16 regions:

Cluster	Size	Top distinguishing terms (TF-IDF)
Mediterranean & Volcanic	16	volcanic, limestone, Mediterranean, indigenous
Southern Hemisphere & International	12	Shiraz, Cabernet, modern, alluvial
French Viticultural	10	Pinot Noir, Chardonnay, premier cru, classified
American West Coast	7	Napa, Sonoma, Cabernet, Zinfandel, valley floor
Iberian Continental	6	Tempranillo, Garnacha, plateau, continental
Germanic Rhine	5	Riesling, slate, steep, river valley
Austrian Danube	3	Grüner Veltliner, loess, terraced

Silhouette score for the terroir clustering is 0.246.

4.3 Independence of the Two Classifications

The central finding is the comparison of the two clusterings:

Statistic	Value
Adjusted Rand Index (ARI)	0.023
Chi-squared statistic (6×7 table)	$\chi^2 = 38.78$
Degrees of freedom	30
Chi-squared p -value	0.131

ARI of 0.023 is statistically indistinguishable from zero; the two clusterings agree no more than two random labellings of 59 regions would be expected to agree. The chi-squared test fails to reject the null hypothesis of independence at the conventional 5% level. By both measures, the identity and terroir classifications are independent.

4.4 Examples of the Independence

The independence is visible at the level of individual regions. Mosel and Pfalz share the Germanic Rhine terroir cluster but belong to different identity clusters (Old World Interior and Old World Exterior, respectively). Burgundy and Northern Rhône are both in the French Viticultural

terroir cluster but, again, belong to different identity clusters (Old World Interior shares them, but their identity neighbours are different). Conversely, Burgundy, Tokaj, and Mosel cluster together by identity (Old World Interior) despite belonging to three different terroir clusters (French Viticultural, Mediterranean & Volcanic, and Germanic Rhine).

5. Robustness Checks

5.1 Sensitivity to k for Identity Clustering

The identity solution is stable for $k \in \{5, 6, 7\}$ in the sense that the six-cluster solution can be recovered as a refinement of the five-cluster and a coarsening of the seven-cluster. ARI between the canonical $k = 6$ solution and the $k = 5$ and $k = 7$ solutions exceeds 0.75. Below $k = 5$, the Old World Exterior and Against the Odds clusters merge, losing meaningful structure. Above $k = 7$, additional clusters split off small subgroups without distinct character.

5.2 Sensitivity to k for Terroir Clustering

The terroir solution is stable for $k \in \{6, 7, 8\}$ with similar ARI overlap. The choice of $k = 7$ maximises silhouette across the candidate range.

5.3 Sensitivity to Encoding Choices

Three alternative pipeline configurations were tested. Stage 1 used Layer 1 TF-IDF only (silhouette 0.114). Stage 2 combined Layer 1 TF-IDF with D-scores at weights $\alpha \in \{0.3, 0.5, 0.7\}$ (silhouette range 0.117–0.135). Stage 3, the chosen design, used D-scores only (silhouette 0.298). The structured expert input dominates the unstructured text; this reinforces the conclusion that identity is a distinct second system rather than a residual of terroir-related vocabulary.

5.4 Permutation Test on the Independence Result

As a non-parametric check on the chi-squared p -value, the cluster labels of one layer were randomly permuted 1,000 times and the ARI recomputed. The observed ARI = 0.023 falls at the 47th percentile of the permutation distribution. This is consistent with independence under a distribution-free test, in agreement with the chi-squared result.

6. Conclusion

6.1 Key Findings

Three findings emerge from the analysis.

First, terroir and identity are statistically independent. A region’s physical environment, as encoded by structured terroir profiles, carries essentially no information about which cultural identity cluster the region belongs to. The Adjusted Rand Index is near zero, and the chi-squared test fails to reject independence.

Second, the independence is structural, not artefactual. It survives changes to k , changes to the encoding pipeline, and a non-parametric permutation test. It is also visible at the level of individual regions: famous identity-similar regions (Burgundy, Tokaj, Mosel) span three different terroir clusters, and famous terroir-similar regions (Mosel, Pfalz, Rheingau) span three identity clusters.

Third, the two systems are interpretable. The identity clusters describe coherent cultural postures (custodial tradition, constructive ambition, etc.); the terroir clusters describe coherent physical environments (Germanic Rhine, Mediterranean & Volcanic, etc.). Each system is internally meaningful; they simply do not align with each other.

The implication for wine regional analysis is precise. Terroir shapes the range of wines a region can produce. Identity — the choice a region makes within that range, expressed through history, ambition, winemaking philosophy, and narrative — operates on a separate axis. Both are real, both are measurable, and they are not the same thing.

6.2 Limitations and Future Work

Three limitations bound the present results. First, the identity dimensions D1–D6 reflect a particular anthropological framing; an alternative framing (e.g., economic posture, generational dynamics) might yield a different identity space. Second, the 59-region selection emphasises canonically recognised regions and underrepresents emerging viticultural areas; a larger sample might surface different cluster structure. Third, the terroir layer is built from English-language source material; regions whose canonical descriptions are richer in their native language may be underspecified.

Future extensions could include: a third layer for wine sensory profile (closing the loop with the TasteRank Explorer framework); a temporal dimension allowing identity drift across vintages or decades; and a cross-validation of the identity dimensions against independent expert panels.

References

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